MATTEO PEZZULLA

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Date of Birth: August 3, 1988
Citizenship: Italian

Summary of qualifications & assets

I am an Associate Professor in the Department of Mechanical Engineering at Aarhus University.

My current research involves aspects of fluid-structure interactions and the theoretical and applied mechanics of thin shells. I work among solid mechanics, fluid mechanics, and differential geometry and I am interested in analytical, numerical, and experimental methods.

I am a member of the American Physical Society (APS).

Experience

June 2025 – currently	Associate Professor at Aarhus University
Nov 2020 – May 2025	Tenure Track Assistant Professor at Aarhus University
Nov 2017 – Oct 2020	Postdoctoral Associate at École Polytechnique Fédérale de Lausanne Supervisor: Prof. Pedro M. Reis
Nov 2015 – Oct 2017	Postdoctoral Associate at Boston University Supervisor: Prof. Douglas P. Holmes

Education

February 2016	Ph.D. in Theoretical and Applied Mechanics, Sapienza - Università di Roma with honors Focus: Morphing of Thin Soft Structures Driven by Geometry and Swelling Advisor: Prof. Paola Nardinocchi
Aug – Dec 2014	Visiting Scholar at Boston University Advisor: Prof. Douglas P. Holmes
October 2012	M.Sc. in Space Engineering, Sapienza - Università di Roma 110/110 cum laude Thesis: "On the control of the large deformations occurring in IPMCs" Advisor: Prof. Paola Nardinocchi
October 2010	B.Sc. in Aerospace Engineering, Sapienza - Università di Roma 110/110 cum laude Thesis: "Bending deformations in ionic polymer metal composites induced by mechano-electro-chemical interactions" Advisor: Prof. Paola Nardinocchi

Invited Seminars

2025 Taming Fluid-Induced Instabilities in Slender Structures for Passive Valve Design. IUTAM Symposium, Manchester, UK, June.

2024 Taming Fluid-Induced Instabilities in Slender Structures for Passive Valve Design. University of Oslo, Oslo, Norway, November.

2024 Taming Fluid-Induced Instabilities in Slender Structures for Passive Valve Design. DTU, Copenhagen, Denmark, October.

2022 Morphing of slender structures via elastic instabilities. CITA, Copenhagen, Denmark, March.

2021 A Geometrically Exact Model for Thin Magneto-elastic Shells. Sapienza University of Rome, Rome, Italy, September.

2021 Natural curvature, pressure, and magnetic fields in shell buckling. Harvard University, Cambridge, USA, February.

2020 Towards Fluid-Shells Interactions. Aarhus University, Aarhus, Denmark, April.

2016 Geometry and Instabilities in Growing Shells. Sapienza - Università di Roma, Rome, Italy, December.

2016 Geometry and Instabilities in Growing Shells. Physical Mathematics Seminar, MIT, Cambridge, MA, October

2015 Morphing of geometric composites. Form Finding Workshop, Roma Tre University, Rome, Italy, April.

2014 Morphing of Geometric Composites via Residual Swelling. Bertoldi Group Meeting, Harvard University, Cambridge, MA, December.

Journals & Conference Proceedings

2025 S. Gennis, M. Pezzulla, and K. H. Jensen. Mechanical limits shape the eccentric form of rose prickles. Submitted (2025)

2025 M. Taffetani and M. Pezzulla. Nonlinear morphoelastic energy based theory for stimuli responsive elastic shells, Available at arXiv:2511.10324. Submitted (2025)

2025 P. G. Ledda, H. Garg, V. Østergaard-Clausen, L. Krumenacker Rudzki, A. Madary, and M. Pezzulla. Fluid-induced snap-through instability of spherical shells, *Phys. Rev. Lett.*, in press (2025)

2024 H. Garg, P. G. Ledda, J. S. Pedersen, and M. Pezzulla. Passive Viscous Flow Selection via Fluid-Induced Buckling. *Phys. Rev. Lett.* 133, 084001 (2024)

2024 K. Barvenik, Z. Coogan, G. Librandi, M. Pezzulla, and E. Tubaldi. Tactile Sensing and Grasping Through Thin-Shell Buckling. *Adv. Intell. Syst.* 2300855 (2024)

2023 Y. L. Lin, M. Pezzulla, and P. M. Reis. Fluid-structure interactions of bristled wings: The trade-off between weight and drag. *J. R. Soc. Interface* 20, 20230266 (2023)

2022 P. G. Ledda, M. Pezzulla, E. Jambon-Puillet, P.-T. Brun, and F. Gallaire. Gravity-driven coatings on curved substrates: a differential geometry approach. *J. Fluid Mech.* 949, A38 (2022)

2022 P. Leroy-Calatayud, M. Pezzulla, A. Keiser, K. Mulleners, and P. M. Reis. Tapered foils favor traveling-wave kinematics to enhance the performance of flapping propulsion. *Phys. Rev. Fluids* 7, 074403 (2022)

2022 M. Pezzulla, D. Yan, and P. M. Reis. A geometrically exact model for thin magneto-elastic shells. *J. Mech. Phys. Solids* 166, 104916 (2022)

2022 T. G. Sano, M. Pezzulla, and P. M. Reis. A Kirchhoff-like theory for hard magnetic rods under three-dimensional geometrically nonlinear deformation. *J. Mech. Phys. Solids* 160, 104739 (2022)

2021 B. Shrimali, M. Pezzulla, S. Poincloux, P. M. Reis, and O. Lopez-Pamies. The remarkable bending properties of perforated plates. *J. Mech. Phys. Solids* 154, 104514 (2021)

2021 D. Yan, M. Pezzulla, L. Cruveiller, A. Abbasi, and P. M. Reis. Magneto-active elastic shells with tunable buckling strength. *Nat. Commun.* 12, 2831 (2021)

2020 M. Pezzulla, E. F. Strong, F. Gallaire, and P. M. Reis. Deformation of porous flexible strip in low and moderate Reynolds number flows. *Phys. Rev. Fluids* 5, 084103 (2020)

2020 D. P. Holmes, J.-H. Lee, H. S. Park, and M. Pezzulla. The nonlinear buckling behavior of a complete spherical shell under uniform external pressure and homogenous natural curvature. *Phys. Rev. E* 102, 023003 (2020)

2020 D. Yan, M. Pezzulla, and P. M. Reis. Buckling of pressurized spherical shells containing a through-thickness defect. *J. Mech. Phys. Solids* 138, 103923 (2020)

2019 M. Pezzulla and P. M. Reis. A Weak Form Implementation of Nonlinear Axisymmetric Shell Equations with Examples. *ASME. J. Appl. Mech.* 86(12): 124502, (2019)

2019 E. F. Strong, M. Pezzulla, F. Gallaire, P. M. Reis, and L. Siconolfi. Hydrodynamic loading of perforated disks in creeping flows. *Phys. Rev. Fluids* 4, 084101, (2019)

2019 A. Lee, D. Yan, M. Pezzulla, D. P. Holmes, and P. M. Reis. Evolution of critical buckling conditions in imperfect bilayer shells through residual swelling. *Soft Matter* 15, 6134-6144, (2019)

2019 L. Stein-Montalvo, P. Costa, M. Pezzulla and D. P. Holmes. Buckling of Geometrically Confined Shells. *Soft Matter* 15, 1215-1222, (2019)

2018 X. Jiang, M. Pezzulla, S. Wei, T. K. Ghosh, and D. P. Holmes. Snapping of Bistable, Prestressed Cylindrical Shells. *Europhys. Lett.* 122, 64003, (2018)

2018 M. Pezzulla, N. Stoop, M. P. Steranka, A. J. Bade, and D. P. Holmes. Curvature-Induced Instabilities of Shells. *Phys. Rev. Lett.* 120, 048002, (2018)

2017 M. Pezzulla, N. Stoop, X. Jiang, and D. P. Holmes. Curvature-Driven Morphing of Non-Euclidean Shells. *Proc. R. Soc. A* 473(2201), 20170087, (2017)

2016 M. Pezzulla, G. P. Smith, P. Nardinocchi, and D. P. Holmes. Geometry and Mechanics of Thin Growing Bilayers. *Soft Matter* 12, 4435-4442, (2016)

2015 P. Nardinocchi, M. Pezzulla, and L. Teresi. Steady and transient analysis of anisotropic swelling in fibered gels. *J. Appl. Phys.* 118, 244904, (2015)

2015 M. Pezzulla, S. A. Shillig, P. Nardinocchi, and D. P. Holmes. Morphing of geometric composites via residual swelling. *Soft Matter* 11, 5812-5820, (2015) **[Inside Front Cover]**

2015 P. Nardinocchi, M. Pezzulla, and L. Teresi, Mechanics of bio-hybrid systems. *Procedia IUTAM* 12, pp. 145-153, (2015)

2015 P. Nardinocchi, M. Pezzulla, and L. Teresi, Anisotropic swelling of thin gel sheets. *Soft Matter* 11, 1492-1499, (2015)

2014 A. Lucantonio, P. Nardinocchi, and M. Pezzulla. Swelling–induced and controlled curving in layered gel beams. *Proc. R. Soc. A* 470(2171), 20140467, (2014)

2014 A. Lucantonio, P. Nardinocchi, M. Pezzulla, and L. Teresi. Multiphysics of bio–hybrid systems: shape control and electro–induced motion. *Smart Mater. Struct.* 23(4), 045043, (2014)

2014 P. Nardinocchi, M. Pezzulla, B.J. Akle, M. Guenther, and T. Wallmersperger. Actuation and buckling effects in IPMCs. *Proc. SPIE* 9056, (2014)

2013 Y. Cha, P. Nardinocchi, M. Pezzulla, and M. Porfiri. Giant displacements in IPMC–based structures: a preliminary study. *Adv. Mat. Res.* 745, 119–128, (2013)

2013 P.Nardinocchi and M. Pezzulla. Curled actuated shapes of ionic polymer metal composites. *J. Appl. Phys.* 113, 224906, (2013)

2011 P. Nardinocchi, M. Pezzulla, and L. Placidi. Thermodynamically based multiphysic modeling of ionic polymer metal composites. *J. Intel. Mat. Syst. Str.* 22(16), 1887–1897, (2011)

Current Group Members

2024- Morten Opstrup Andersen, PhD Student (HFSP project)

Honors & Grants

2023 HFSP, Research grant - Program, co-PI (USD 1.5M),

2023 The Royal Society, International Exchanges, co-PI (GBP 5k),

2022 VILLUM Experiment, VILLUM Fonden, PI (DKK 2M),

2022 Thomas B. Thriges Fond, PI (DKK 40k),

2016 PHASME 2016 Travel Grant, ICAM (USD 1k),

2014 Research Project Grant *Giovani Ricercatori*, INdAM (€ 1.6k), "Corrugamento di travi bistrato di gel polimerico"

2014 Research Project Young Investigator Grant, Sapienza University (€ 2k), "Shaping of bio-hybrid systems: reduced models and numerical simulations"

2013–2015 Graduate Research Fellowship, Italian Ministry of Education

2007–2012 ADISU scholarship

Teaching, Teaching Qualifications, and Service

2024 Co-organizer of the 58th meeting of the Society for Natural Philosophy, Aarhus

2024 Co-organizer of the Symposium: Mechanics of Soft Materials and Structures, at EMMC19, Madrid

2024 Foundational course in PhD Supervision, Diploma

2021- Courses: *Slender Structures*, *Theory of Elasticity*
AU, Instructor

2021 University Pedagogical Programme at AU, Diploma

2021- Member of the Education Committee at AU

2019 Lectures on shells for the class on *Mechanics of Slender Structures*
EPFL, Instructor: Prof. Pedro M. Reis

Language & Programming Skills

- ⋆ Mother tongue: Italian. Fluent in English and French. B2 in Spanish and Danish.
- ⋆ Experience programming in C, Fortran, and Python.
- ⋆ Experience with mathematical software such as Matlab, Mathematica, LabView, LaTeX.
- ⋆ Experience with FE software such as COMSOL Multiphysics, ADINA, Nastran.
- ⋆ Experience with Adobe Illustrator, Blender, MeshLab, Paraview and ImageJ.

November, 2025